

VMS Workstation Server

Final Build — Bill of Materials

150-camera Panasonic i-PRO PI-BUA3MWA | 1080p H.265 | 5-day retention | 300 concurrent users | AI inference enabled

Cameras	Per-cam daily	Total per day	5-day retention need	Provisioned usable
150	150 GB	22.5 TB	~155 TB (with 80% safe-fill)	~217 TB (28% headroom)

Workload profile. Recording driven by shift-code (recording active only while shift code is running — 150 GB/day per camera is the real measured output). Bulk archive on enterprise SATA HDD via ZFS RAID-Z2 with NVMe hot tier for last-24h footage and AI working set. Dual GPU for decode redundancy plus AI inference headroom for future face / ANPR / multi-camera tracking.

Decisions locked-in for this BoM:

Decision	Pick	Rationale
Chassis form	Supermicro 740GP-TNRT (rack-tower hybrid)	IPMI BMC for remote console + hot-swap drive bays. Plant 24/7 deployment cannot afford taking the box down to swap a failed disk.
CPU tier	AMD Threadripper PRO 7975WX (32C / 64T)	+ 8 cores over 7965WX for AI pipeline + 300-user fan-out headroom. CPU upgrade later costs 4× what choosing right now does.
Bulk drive	12× Seagate Exos X20 (20 TB, CMR/PMR)	5+ years field data, 550 TB/year workload rating. X22 (HAMR) is newer-tech with less plant-deployment proof. 20 TB already yields ~7-day actual retention.

1. Bill of Materials

Prices indicative for India, FY 2026, GST inclusive. Vendor quotes will vary $\pm 5\%$ based on stock + import currency.

#	Component	Part / Model	Qty	Unit ?	Total ?
1	CPU	AMD Ryzen Threadripper PRO 7975WX (32C/64T, 5.3 GHz)	1	6,50,000	6,50,000
2	Motherboard	ASUS Pro WS WRX90E-SAGE SE (sTR5, 8-ch DDR5 RDIMM)	1	1,50,000	1,50,000
3	RAM	Kingston KSM52R42BD8-32MD — 32 GB DDR5-5200 ECC RDIMM	8	22,000	1,76,000
4	GPU	NVIDIA L4 24 GB (PG194, single-slot 72 W)	2	3,00,000	6,00,000
5	Boot SSD	Samsung 990 PRO 2 TB M.2 NVMe PCIe 4.0	2	20,000	40,000
6	Hot-tier SSD	Kioxia CD8-R 7.68 TB U.2 NVMe (1 DWPD enterprise)	4	1,75,000	7,00,000
7	Bulk-tier HDD	Seagate Exos X20 20 TB SATA 7200 RPM (ST20000NM007D)	12	50,000	6,00,000
8	HBA	Broadcom LSI 9500-16i (PCIe 4.0, 16-port, IT mode)	1	70,000	70,000
9	Chassis	Supermicro SuperWorkstation 740GP-TNRT (4U tower/rack)	1	1,50,000	1,50,000
10	PSU	Supermicro PWS-1K63A-1R 1600 W Platinum (CRPS, redundant)	2	45,000	90,000
11	CPU cooler	Noctua NH-U14S TR5-SP6 (air, 240 W TDP)	1	12,000	12,000
12	Case fans	Noctua NF-A12x25 PWM 120 mm	6	3,000	18,000
13	SAS / SATA cables	Mini-SAS HD $\times$ 4 (SFF-8643 to SFF-8643, 1 m)	4	2,500	10,000
14	UPS	APC Smart-UPS SRT 3000 VA online + Network MGMT card	1	1,20,000	1,20,000
		HARDWARE TOTAL			~33,86,000

Note on Hot-tier SSD (line 6). Kioxia CD8-R is the U.2 NVMe enterprise drive with 1 DWPD endurance — deliberate over-spec versus consumer NVMe because the hot tier absorbs every camera's first-write before ZFS pushes it to the HDD pool. Average write load on this tier $\approx$ 30 MB/s sustained per drive, well within endurance budget for 5-year life.

2. Storage Layout

Tier	Drives	Config	Raw	Usable	Role
OS	2× 2 TB Samsung 990 PRO M.2 NVMe	mdraid RAID 1	4 TB	2 TB	Ubuntu 24.04 LTS, Postgres data dir, Frigate config + DB, system journal
Hot	4× 7.68 TB Kioxia CD8-R U.2 NVMe	mdraid RAID 10	30.7 TB	15.36 TB	Write-hot zone: incoming RTSP frames, last-24 h footage, AI working set, snapshot index
Bulk	12× 20 TB Seagate Exos X20 SATA	ZFS RAID-Z2 (10+2)	240 TB	~200 TB	5-day archive (need 155 TB → ~7-day actual retention margin)
		GRAND TOTAL USABLE		~217 TB	

2a. ZFS pool configuration (apply on first boot)

```
zpool create -o ashift=12 tank raidz2 \  
/dev/sd[b-m] # 12 Exos X20 drives  
zfs set recordsize=1M tank # video files love 1M records  
zfs set compression=lz4 tank # ~5-8% saving on metadata  
zfs set atime=off tank # no last-access updates  
zfs set xattr=sa tank # extended attr inside dnode  
zfs set redundant_metadata=most tank # extra metadata parity  
echo "options zfs zfs_arc_max=206158430208" \  
> /etc/modprobe.d/zfs.conf # ARC cap = 192 GB (75% of RAM)  
systemctl enable zfs-scrub-weekly@tank # Sunday 02:00 weekly scrub
```

Why no L2ARC / SLOG? Hot-tier NVMe RAID-10 already plays the role of write-cache for incoming footage; ARC sits at 192 GB in RAM for read-cache. Adding L2ARC would only displace ARC RAM tracking. SLOG is for synchronous writes which the recording pipeline does not issue.

2b. Retention math (verification)

Step	Calculation	Result
Daily ingest (150 cams)	150 GB × 150	22.5 TB / day
5-day raw archive	22.5 TB × 5	112.5 TB
+ index/metadata/audit (10%)	× 1.10	~124 TB
+ safe-fill headroom (80% max)	÷ 0.80	~155 TB target
Provisioned ZFS pool	12× 20 TB RAID-Z2	~200 TB usable
Headroom over target	(200 – 155) / 155	+29% — comfortable
Actual retention at 22.5 TB/day	200 ÷ 22.5	~8.8 days real

3. Compute & Decode Capacity Verification

Demand	Pipeline element	Engineering check	Verdict
150 RTSP demux + 300 user fan-out	Threadripper PRO 7975WX 32C/64T	Frigate worker pool $\approx$ 1 thread per camera plus nginx workers; 32C handles 150 + 60 nginx + OS = < 75% load	OK — 25%+ headroom
256 GB RAM	ZFS ARC 192 GB + Frigate 25 GB + Postgres 16 GB + OS 10 GB	Sum $\approx$ 243 GB peak. RAM ECC scrubbing handled in background.	OK — 5% spare
150 $\times$ 1080p H.265 decode @ ~14 Mbps	2 $\times$ NVIDIA L4 (8 NVDEC + 4 NVENC total)	Single L4 = 130 streams ceiling; pair gives 260-stream ceiling + fault tolerance on single-GPU failure	OK — 73% headroom
AI inference (YOLOv8 @ 1-5 fps $\times$ 150 streams)	L4 484 INT8 TOPS each $\times$ 2	YOLOv8n at 1 fps $\times$ 150 streams = ~600 inferences/sec — single L4 ceiling is ~1500/sec	OK — room for 2nd AI model
22.5 TB/day write — bulk pool	12-drive RAID-Z2, ZFS	Daily write $\approx$ 260 MB/s sustained avg; 12-disk RAID-Z2 sequential = 600-800 MB/s	OK — 2.5$\times$ headroom
~5 min UPS runtime	APC SRT 3000 VA, 700 W actual draw	Battery rated 5-10 min at 700 W. apcupsd signals graceful shutdown at 50% battery to avoid ZFS journal corruption	OK — survives short outages, clean shutdown on long

Bottleneck analysis. The first ceiling to hit (if scale increases) is **RAM** — growth path is to bump 8 $\times$ 32 GB $\rightarrow$ 8 $\times$ 64 GB = 512 GB (~+1.8 lakh). Storage bottleneck is much further: 8 drive bays remain free in the 740GP chassis for +160 TB expansion (= +5 days retention) at any time.

4. Software Stack (Recommended)

Layer	Pick	Why
Operating system	Ubuntu Server 24.04 LTS	NVIDIA + ZFS + i-PRO ONVIF all officially supported. 5-year LTS to 2029.
Filesystem (bulk)	ZFS RAID-Z2 with lz4 compression	Inline checksums catch HDD bit-rot; weekly scrub verifies redundancy; snapshots for instant rollback if footage is accidentally deleted.
NVR / VMS core	Frigate NVR (open source) + MediaMTX	GPU-accelerated decode pipeline; native i-PRO ONVIF probe; MQTT events for downstream integration with MES.
AI pipeline	NVIDIA DeepStream 7 + YOLOv8 (ByteTrack for person re-identification)	Purpose-built for video AI on NVIDIA hardware; pulls RTSP directly, GPU end-to-end with zero CPU copies.
Browser delivery	go2rtc — WebRTC sub-second relay	Serves sub-stream to 300 user browsers without transcoding; lowest latency for live view.
Reverse proxy	nginx + Let's Encrypt TLS	HTTPS termination + URL routing + connection pool to backend.
Monitoring	Prometheus + Grafana + Loki	Tracks: GPU temp & NVDEC utilisation, HDD SMART (predict failure), ZFS pool health, RTSP packet drops per camera, dropped frames count.
Backup (cold)	Restic → NAS or AWS S3 Glacier (event clips only)	Critical incident clips replicated off-box; not full archive.
Camera integration	ONVIF Profile S/T probe + i-PRO Smart Coding enabled	Lowers bitrate during static frames (your factory bullet cameras can drop from 14 Mbps to 6-8 Mbps in low-motion windows).

5. Install / Commissioning Checklist

#	Action	Notes
1	Receive & serial-number check	Match each component against this BoM; photograph each drive serial for warranty.
2	BIOS / firmware update	ASUS WRX90E to latest BIOS, Supermicro IPMI to latest, all 12 HDDs to latest Seagate fw (critical — old fw has known firmware bugs on 20 TB SKU).
3	Memory burn-in	Memtest86 24-hour pass before OS install; ECC errors logged.
4	OS install + ZFS module + driver pin	Ubuntu 24.04 LTS minimal install, NVIDIA 550-series driver pinned via apt-mark hold.
5	ZFS pool creation & tuning	Apply section 2a commands; create first snapshot baseline.
6	Hot-tier RAID + mountpoint	mdadm RAID 10 across 4 Kioxia NVMe; mount at /var/lib/frigate/hot.
7	NVIDIA driver + DeepStream + CUDA	Verify with nvidia-smi: 2 L4 detected, NVDEC available.
8	Frigate / MediaMTX / nginx deploy via Docker Compose	Single compose file, image-tag pinned; volumes mounted to hot & bulk tiers.
9	Add 1st camera, verify pipeline end-to-end	RTSP probe → decode → record → ZFS write → browser playback → AI event → MQTT.
10	Roll out remaining 149 cameras in batches of 10	Watch ZFS write latency & GPU NVDEC util per batch; no batch should push utilisation past 60%.
11	Monitoring stack live	Prometheus targets up, Grafana dashboards loaded, Loki collecting Frigate logs, AlertManager wired to email / Slack.
12	UPS & graceful-shutdown rehearsal	Pull mains plug while system is live; verify apcupsd triggers clean shutdown before battery exhausts.
13	DR drill	Pull one bulk-tier disk; verify ZFS reports DEGRADED state; replace; verify resilver completes; pull second disk; verify still online.
14	User load test	Apache Bench / wrk simulate 300 concurrent browser clients pulling sub-stream; verify no frame drops, CPU < 60%, GPU NVDEC < 50%.

6. Power, Thermal & Physical

Item	Value
Idle draw (cameras connected, no recording)	~280 W
Typical operating draw (recording + AI)	~650 W
Worst-case sustained (heavy AI + ZFS scrub)	~850 W
PSU rating (Platinum, redundant 1+1)	1600 W × 2 = 1600 W usable
UPS rating (online, 3 kVA)	Holds ~10 min at 700 W draw
Heat dissipation worst-case	~2900 BTU/hr
Rack-room ambient max (sustained)	< 30 °C inlet — ideal 22-24 °C
Chassis form-factor	4U tower / rack convertible (Supermicro 740GP)
Chassis weight populated	~38 kg (12 HDD + 2 GPU + CPU + PSU)
Acoustic level (rack mode, full load)	~55 dBA at 1 m — not silent, rack-room class
Acoustic level (tower mode, typical load)	~42 dBA at 1 m — office-tolerable

Cooling recommendation. Install in a dedicated rack room with active extraction or AC dropping inlet temperature to 22-24 °C. The Supermicro 740GP has front-to-back airflow with hot-aisle exhaust; pair with closed-door 18U/24U rack if more equipment will join later.

Annual energy estimate. Average draw 650 W × 24 h × 365 d = 5694 kWh/yr. At 8 INR/kWh industrial tariff = **~45,500 INR/year** running cost.

7. Future Expansion Path (locked-down room for growth)

Future need	Add	Cost (INR lakh)
Retention → 10 days (instead of 5)	+6× 20 TB Exos X20 (drive bays already in chassis)	+3.0
Retention → 14 days	External JBOD shelf (Supermicro 847E2C) + 12 more 20 TB drives	+9.5
Cameras → 250+	RAM upgrade 256 → 512 GB; existing GPUs handle decode	+1.8
4K migration on camera fleet	Swap 2× L4 for 1× RTX 6000 Ada (48 GB) — better 4K decode	+5.0 (net)
Face recognition / ANPR / multi-cam tracking	Add 2nd RTX 4000 Ada or keep 2× L4 and load 2nd AI model	+0.0 to +4.0
Hot-DR / second site replication	ZFS send/recv to identical box at second location	Clone build + pipe

8. Vendor & Procurement Notes

Line item	Suggested sourcing
CPU + Motherboard	Ingram Micro India / Rashi Peripherals authorised ASUS + AMD reseller. TR PRO 7975WX has 6-week lead time globally; pre-book.
NVIDIA L4	PNY / Leadtek through Avnet India; demand letter via NVIDIA NPN partner. Stock-tight SKU — quote-confirm before ordering balance of build.
Seagate Exos X20 × 12	Spear Technologies / Iris Global (Seagate AAP). Insist on matched batch (consecutive serials) to avoid mixed-firmware ZFS edge cases.
Kioxia CD8-R NVMe	Acme India / Compuage Infocom. Stock through datacenter SKUs — budget 8-week lead if ordering from Singapore stock.
Supermicro 740GP	Direct via Supermicro India (Bangalore) or NetWeb Technologies. Specify TPM 2.0 module + rail kit at order time.
APC SRT 3000	Schneider Electric India direct. Always order with battery upgrade option for +15 min runtime if budget allows.
Warranty & AMC	Negotiate 5-year on-site NBD (next-business-day) warranty for: chassis, PSU, motherboard, GPU, HDD. CPU has 3-year manufacturer; RAM lifetime.

9. Final Cost Summary

Category	INR
Compute (CPU + board + RAM)	9,76,000
Graphics (2× L4)	6,00,000
Storage (OS + Hot + Bulk)	13,40,000
HBA + cables	80,000
Chassis + PSU + cooling	2,70,000
UPS	1,20,000
HARDWARE TOTAL	~33,86,000
Software / OS (FOSS stack)	0
Install & commissioning (1 week, 2 engineers)	1,50,000
Annual AMC (Year 2 onwards, 5%)	1,70,000/yr
YEAR-1 LANDED COST	~35,36,000

Document prepared for Toyota Boshoku Device India — EOL MES project, VMS expansion phase. All hardware specs validated for 1080p H.265, 150 cameras, 5-day retention, 300 concurrent users, AI inference enabled. Engineering rationale documented per component. Budget hold-back recommended: 5% of hardware total (~1.7 lakh) for accessory and freight overruns.